

UNIT 5 · NON-LINEAR STRUCTURES

16. Trees and General Trees

CSC211 · Algorithms and Data Structures

COMSATS University Islamabad, Lahore Campus

Agenda

N

Why Non-Linear Structures?

One element can branch into many children at once

B

Binary Trees

At most two children — full, complete, and skewed shapes

G

General Tree -> Binary Tree

The first-child/next-sibling (FCNS) trick

T

Tree Concept & Terminology

Root, parent, child, sibling, leaf, depth, height

[]

Array vs. Linked Representation

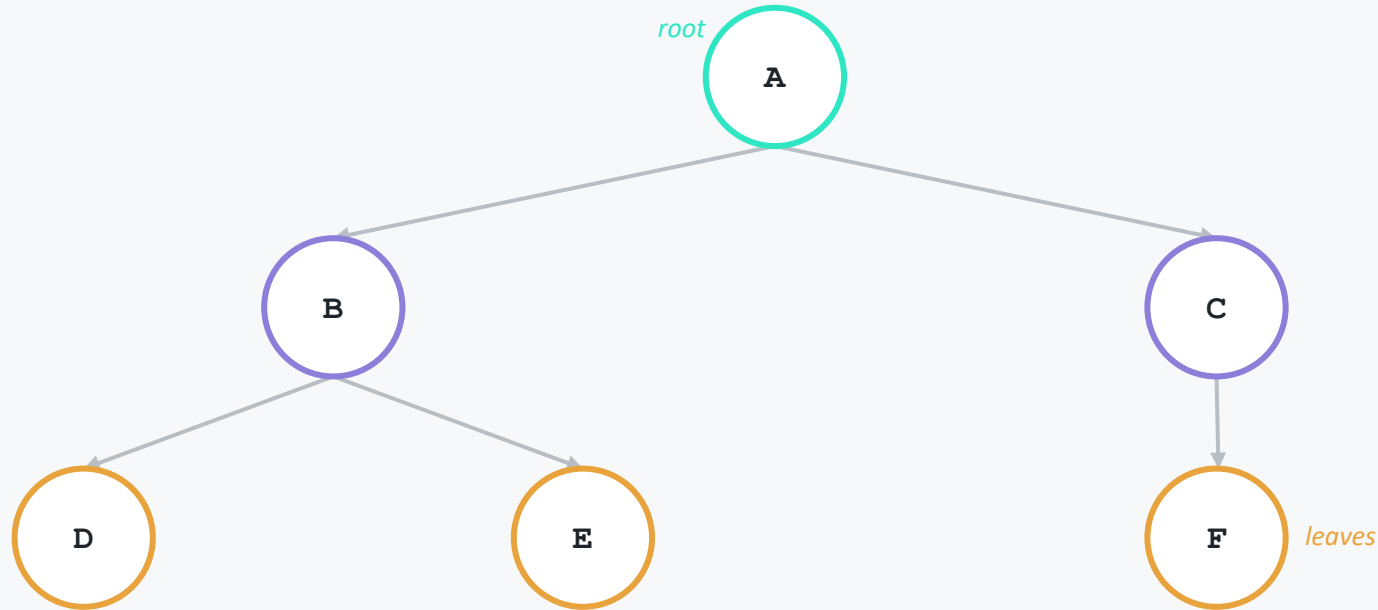
Two ways to store a binary tree in memory

W

Real-World Applications

File systems, org charts, the DOM, compilers

The Tree Concept and Terminology



Root

No parent — the top (A)

Leaf

No children (D, E, F)

Sibling

Same parent (B, C)

Depth

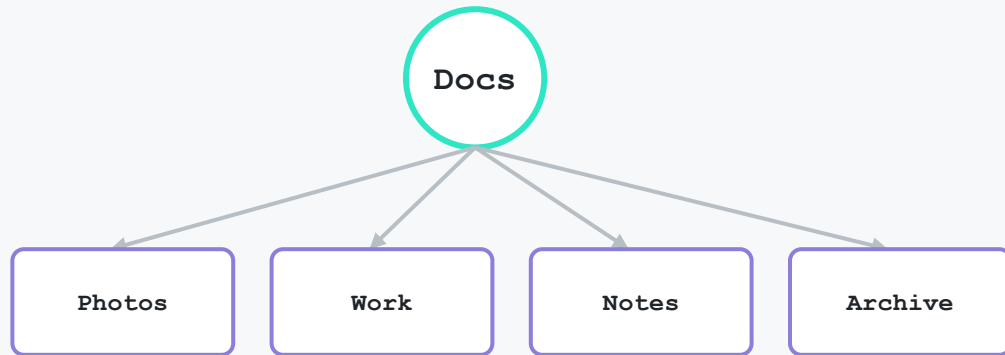
Edges from root down

Height

Longest root-to-leaf path (2)

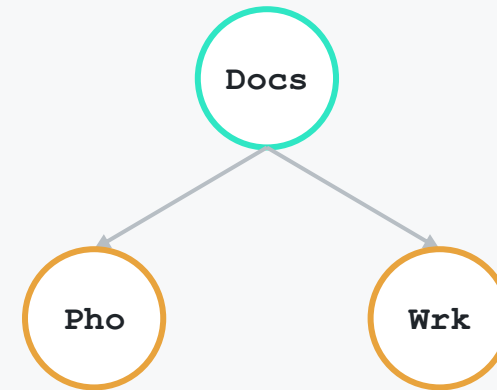
A general tree places no limit on how many children a node can have — binary trees add one restriction to simplify analysis.

General Tree vs. Binary Tree



General tree — any number of children

Documents has FOUR children — perfectly normal for a folder.



Binary tree — at most 2 children

No third or fourth pointer to use — needs a different strategy entirely.

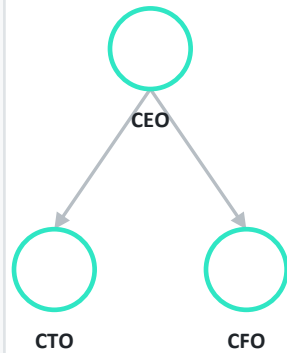
This isn't cosmetic: it's the reason binary trees need a completely different strategy (first-child/next-sibling, later this lecture) to represent a general tree's unlimited branching at all.

Building a Tree, Node by Node

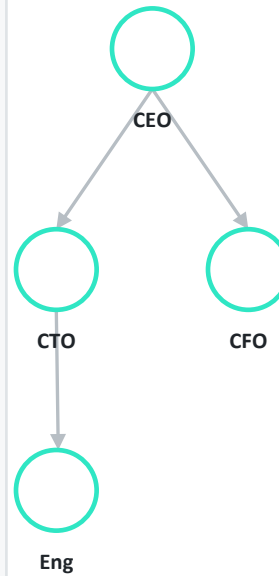
Step 1



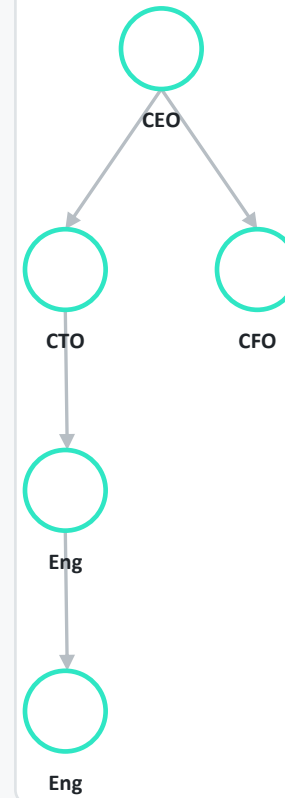
Step 2: + CTO, CFO



Step 3: + Eng Lead

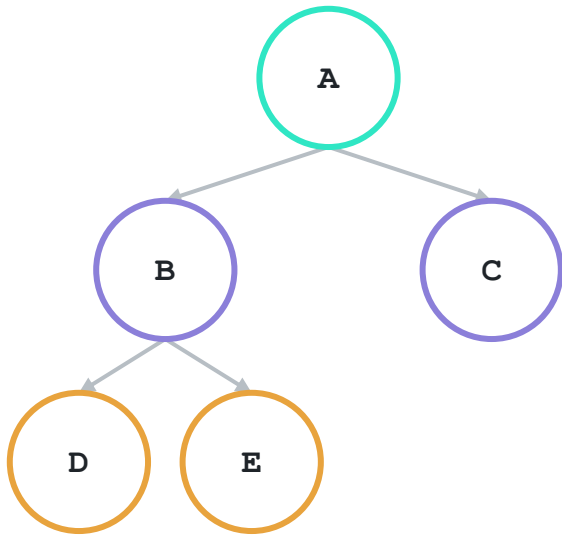
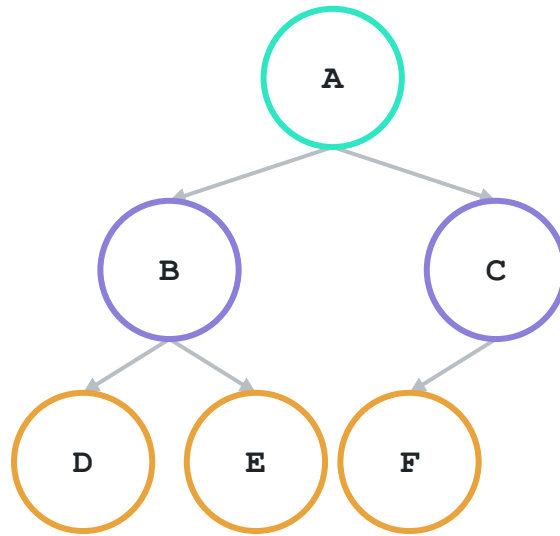
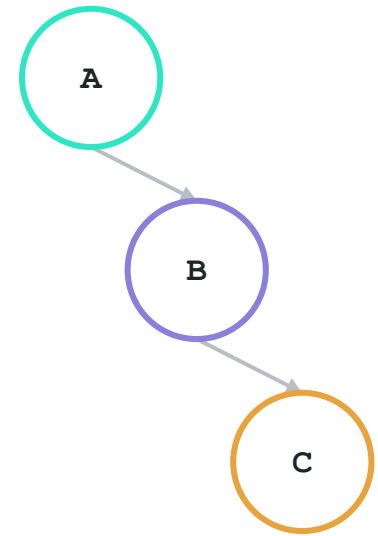


Step 4: + Engineer



Height after Step 4: 3 edges from root (CEO) to deepest leaf (Engineer).

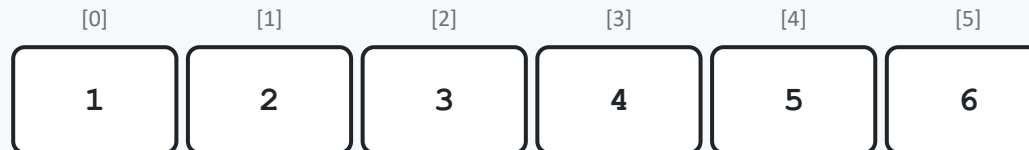
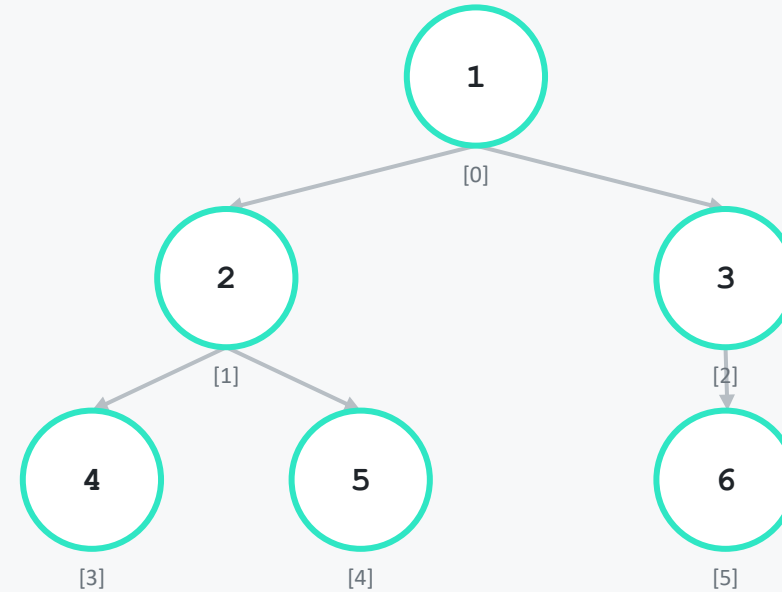
Full, Complete, and Skewed Binary Trees

Full**Complete****Skewed**

Full: every node has 0 or 2 children. Complete: every level full except possibly the last, left to right. Skewed: every node has one child — the worst case for search (Lecture 21).

Sequential (Array) Representation

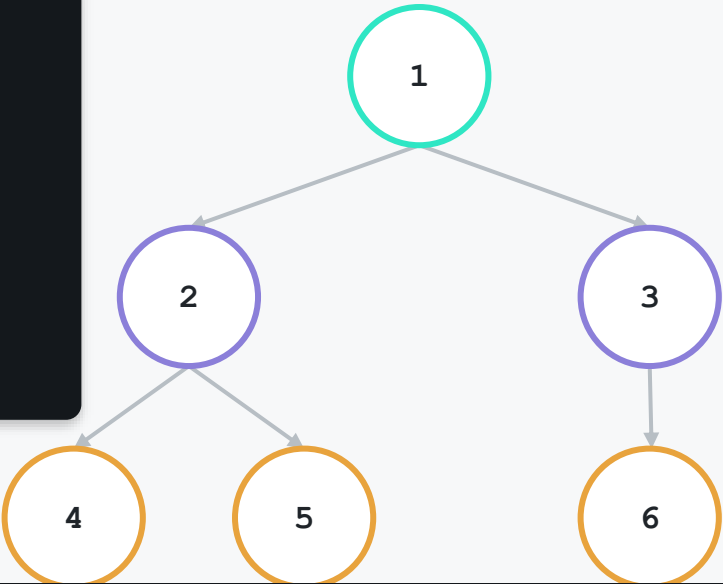
```
index arithmetic  
left  = 2*i + 1  
right = 2*i + 2  
parent = (i - 1) / 2
```



Compact and cache-friendly — but wastes space badly for a skewed (incomplete) tree.

Linked Representation of Binary Trees

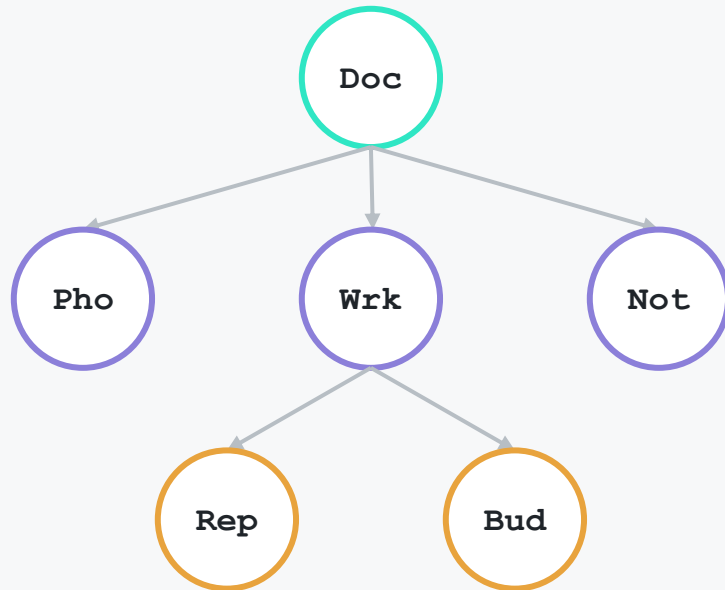
```
TreeNode
struct TreeNode {
    int data;
    TreeNode* left;
    TreeNode* right;
    TreeNode(int v) : data(v),
        left(nullptr), right(nullptr) {}
};
```



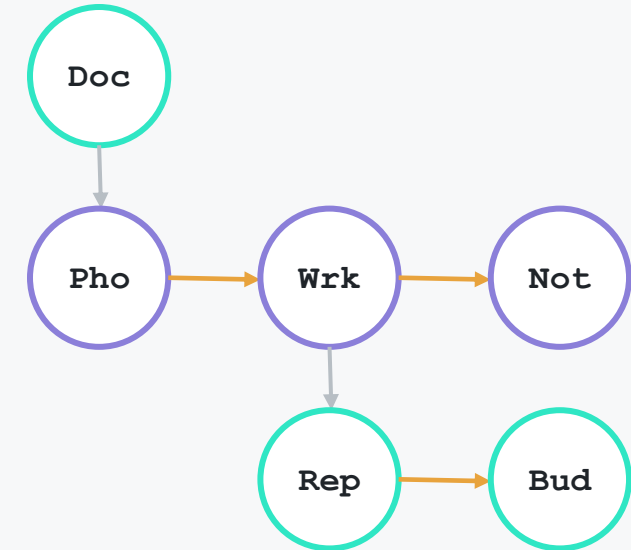
	Array Representation	Linked Representation
Best for	Complete trees (heaps)	Any binary tree shape
Memory	Compact, wastes on gaps	One node + 2 pointers each
Finding a child	O(1) arithmetic	O(1) pointer dereference

First-Child / Next-Sibling (FCNS)

left = this node's FIRST CHILD. right = this node's NEXT SIBLING.



General tree (Documents: 3 children)



FCNS binary encoding (down=firstChild, right=nextSibling)

The FCNS Node and Conversion

`general_to_binary.cpp`

```
struct FCNSNode {
    string name;
    FCNSNode* firstChild;    // was "left"
    FCNSNode* nextSibling;  // was "right"
};

FCNSNode* convert(GeneralNode* node) {
    FCNSNode* converted = new FCNSNode(node->name);
    converted->firstChild = convert(node->children[0]);
    FCNSNode* sibling = converted->firstChild;
    for (size_t i = 1; i < node->children.size(); i++) {
        sibling->nextSibling = convert(node->children[i]);
        sibling = sibling->nextSibling;
    }
    return converted;
}
```

Real-World Applications of Trees

File Systems

A folder can hold any number of items — navigation is naturally hierarchical.

Org Charts

A manager's "next" isn't one person — it's a whole team of direct reports.

The DOM

A `<div>` can contain any number of children — rendering & events walk this tree.

Decision Trees

Each answer branches to a completely different next question.

Syntax Trees

$a + b * c$ nests $b * c$ inside the addition — the structure IS a tree.

KEY TAKEAWAYS

Key Takeaways

- A tree is a hierarchical, non-linear structure — one node can have multiple children, unlike every linear structure so far.
- Core terminology — root, parent, child, sibling, leaf, depth, height, subtree — is used throughout the rest of this unit.
- A binary tree restricts every node to at most two children; it can be full, complete, or skewed, which affects search efficiency.
- The array representation is compact but only efficient for complete trees; the linked representation handles any shape.
- A general tree can be represented with ordinary two-pointer nodes via first-child/next-sibling (FCNS) encoding.